Lab0: GDB&QEMU 调试64位RISC-V LINUX

实验目的：

- 使用交叉编译工具，完成Linux内核代码编译
- 使用QEMU运行内核
- 熟悉GDB和QEMU联合调试

实验过程与代码实现

搭建实验环境

> 实验在vmware虚拟机上运行

1. 确定当前linux版本：运行`lsb_release -a`和`uname -m`命令，

```
dahyunnist@dahyunnist-ubuntu:~$ lsb_relaese -a
Command 'lsb_relaese' not found, did you mean:
  command 'lsb_release' from deb lsb-release (12.0-2)
Try: sudo apt install <deb name>
dahyunnist@dahyunnist-ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
dahyunnist@dahyunnist-ubuntu:~$ uname -m
x86_64
```

   查看当前linux版本为Ubuntu 24.04， 架构为x86_64，符合实验要求：
2. 更新软件包列表以确保安装最新的软件包：
   - 运行命令`sudo apt update`：
   - 运行命令`sudo apt upgrade -y`升级所有可升级软件包：
3. 安装编译内核所需要的交叉编译工具链和用于构建程序的软件包：
   - 运行命令`sudo apt install gcc-riscv64-linux-gnu`安装RISC-V 64位架构的交叉编译器
   - 安装所需的构建工具链，数学与底层库和辅助工具:

```
sudo apt install  autoconf automake autotools-dev curl libmpc-dev
libmpfr-dev libgmp-dev \
gawk build-essential bison flex texinfo gperf libtool patchutils bc \
zlib1g-dev libexpat-dev git
```

4. 安装用于启动riscv64平台上的内核的模拟器`qemu`
   - 运行命令`sudo apt inatsll qemu-system-misc`:
5. 安装多架构`gdb`用于对在`qemu`上运行的Linux内核进行调试：

- 运行命令`sudo apt install gdb-multiarch`:

```
dahyunnist@dahyunnist-ubuntu:~$ sudo apt install gdb-multiarch
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  gdb-multiarch
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 5,152 kB of archives.
After this operation, 20.7 MB of additional disk space will be used.
Get:1 http://mirrors.tuna.tsinghua.edu.cn/ubuntu noble/universe amd64 gdb-multiarch amd6
4 15.0.50.20240403-0ubuntu1 [5,152 kB]
Fetched 5,152 kB in 1s (3,615 kB/s)
Selecting previously unselected package gdb-multiarch.
(Reading database ... 220681 files and directories currently installed.)
Preparing to unpack .../gdb-multiarch_15.0.50.20240403-0ubuntu1_amd64.deb ...
Unpacking gdb-multiarch (15.0.50.20240403-0ubuntu1) ...
Setting up gdb-multiarch (15.0.50.20240403-0ubuntu1) ...
```

获取Linux源码和已经编译好的文件系统

1. 从 https://www.kernel.org 下载最新的 Linux 源码，放置在`/home/dahyunnist/os/linux`路径下
2. 克隆课程仓库至`/home/dahyunnist/os`路径下并验证根文件系统镜像(`os-25fall/src/la0/rootfs.img`):

```
dahyunnist@dahyunnist-ubuntu:~/os$ git clone https://git.zju.edu.cn/zju
-os-sld/os-25fall
Cloning into 'os-25fall'...
warning: redirecting to https://git.zju.edu.cn/zju-os-sld/os-25fall.git
/
remote: Enumerating objects: 250, done.
remote: Counting objects: 100% (185/185), done.
remote: Compressing objects: 100% (176/176), done.
remote: Total 250 (delta 95), reused 42 (delta 4), pack-reused 65 (from
 1)
Receiving objects: 100% (250/250), 9.22 MiB | 8.82 MiB/s, done.
Resolving deltas: 100% (99/99), done.
dahyunnist@dahyunnist-ubuntu:~/os$ ls
linux  os-25fall
dahyunnist@dahyunnist-ubuntu:~/os$ cd os-25fall/src/lab0
dahyunnist@dahyunnist-ubuntu:~/os/os-25fall/src/lab0$ ls
rootfs.img
```

编译Linux内核

1. 使用默认配置：

```
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$ make ARCH=riscv C
ROSS_COMPILE=riscv64-linux-gnu- defconfig
  HOSTCC  scripts/basic/fixdep
  HOSTCC  scripts/kconfig/conf.o
  HOSTCC  scripts/kconfig/confdata.o
  HOSTCC  scripts/kconfig/expr.o
  LEX     scripts/kconfig/lexer.lex.c
  YACC    scripts/kconfig/parser.tab.[ch]
  HOSTCC  scripts/kconfig/lexer.lex.o
  HOSTCC  scripts/kconfig/menu.o
  HOSTCC  scripts/kconfig/parser.tab.o
  HOSTCC  scripts/kconfig/preprocess.o
  HOSTCC  scripts/kconfig/symbol.o
  HOSTCC  scripts/kconfig/util.o
  HOSTLD  scripts/kconfig/conf
*** Default configuration is based on 'defconfig'
#
# configuration written to .config
#
```

2. 运行命令`make ARCH=riscv CROSS_ComPILE=riscv64-linux-gnu- -j$(nproc)`开始编译，编译后生成内核镜像：

```
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$ ls arch/riscv/boot/Image
arch/riscv/boot/Image
```

使用QEMU运行内核

1. 运行命令：

```
$ qemu-system-riscv64 \
   -nographic \
   -machine virt \
   -kernel ~/os/linux/linux-6.16.9/arch/riscv/boot/Image \
   -device virtio-blk-device,drive=hd0 \
   -append "root=/dev/vda ro console=ttyS0" \
   -bios default \
   -drive file=~/os/os-25fall/src/lab0/rootfs.img,format=raw,id=hd0 #使用
lab0/rootfs.img作为根文件系统
```

```
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$ qemu-system-riscv64 -nographic -machine virt
 -kernel ~/os/linux/linux-6.16.9/arch/riscv/boot/Image \
> -device virtio-blk-device,drive=hd0 -append "root=/dev/vda ro console=ttyS0" \
> -bios default -drive file=~/os/os-25fall/src/lab0/rootfs.img,format=raw,id=hd0

OpenSBI v1.3
   ____                    _____ ____ _____
  / __ \                  / ____|  _ \_   _|
 | |  | |_ __   ___ _ __ | (___ | |_) || |
 | |  | | '_ \ / _ \ '_ \ \___ \|  _ < | |
 | |__| | |_) |  __/ | | |____) | |_) || |_
  \____/| .__/ \___|_| |_|_____/|___/_____|
        | |
        |_|

Platform Name             : riscv-virtio,qemu
Platform Features         : medeleg
Platform HART Count       : 1
Platform IPI Device       : aclint-mswi
Platform Timer Device     : aclint-mtimer @ 10000000Hz
Platform Console Device   : uart8250
Platform HSM Device       : ---
Platform PMU Device       : ---
Platform Reboot Device    : sifive_test
```

2. 使用`uname -a`命令查看当前架构的确为riscv64：

```
[   12.279668] EXT4-fs (vda): mounted filesystem c3e9bbca-ec22-47f9-a368-187b21172fc1 ro with orde
red data mode. Quota mode: disabled.
[   12.289398] VFS: Mounted root (ext4 filesystem) readonly on device 254:0.
[   12.306024] devtmpfs: mounted
[   12.642842] Freeing unused kernel image (initmem) memory: 2388K
[   12.648429] Run /sbin/init as init process

Please press Enter to activate this console.
/ # uname -a
Linux (none) 6.16.9 #1 SMP Fri Sep 26 00:21:35 CST 2025 riscv64 GNU/Linux
/ #
```

使用GDB对内核进行调试

1. 终端1启动qemu，执行如下命令： 运行命令：

```
$ qemu-system-riscv64 \
   -nographic \
   -machine virt \
   -kernel ~/os/linux/linux-6.16.9/arch/riscv/boot/Image \
   -device virtio-blk-device,drive=hd0 \
   -append "root=/dev/vda ro console=ttyS0" \
   -bios default \
   -drive file=~/os/os-25fall/src/lab0/rootfs.img,format=raw,id=hd0 -S -s
```

   其中`-S`参数表示**启动时暂停CPU执行**，`-s`是`-gdb tcp::1234`的简写，表示**在1234端口启动gdb服务器，等待调试连接。**

2. 终端2启动gdb并连接
   - 执行`gdb-multiarch ~/os/linux/linux-6.16.9/vmlinux`
   - 启动后执行命令`target remote :1234`，连接qemu的gdb服务器（端口1234）：

```
(gdb) target remote :1234
Remote debugging using :1234
0x0000000000001000 in ?? ()
```

- 执行命令`b start_kernel`，在`kernel/init/main.c`的`start_kernel`函数设置断点：

```
(gdb) b start_kernel
Breakpoint 1 at 0xffffffff80c006b0
```

- 执行命令`continue`，两终端输出如下：

```
dahyunnist@dahyunnist-ubuntu: ~/os/linux/linux-6.16.9
Firmware RW Offset        : 0x40000
Firmware RW Size          : 66 KB
Firmware Heap Offset      : 0x48000
Firmware Heap Size        : 34 KB (total), 2 KB (reserved), 9 KB (used), 22 KB (f
ree)
Firmware Scratch Size     : 4096 B (total), 760 B (used), 3336 B (free)
Runtime SBI Version       : 1.0

Domain0 Name              : root
Domain0 Boot HART         : 0
Domain0 HARTs             : 0*
Domain0 Region00          : 0x0000000002000000-0x000000000200ffff M: (I,R,W) S/U:
 ()
Domain0 Region01          : 0x0000000080040000-0x000000008005ffff M: (R,W) S/U: (
)
Domain0 Region02          : 0x0000000080000000-0x000000008003ffff M: (R,X) S/U: (
)
Domain0 Region03          : 0x0000000000000000-0xffffffffffffffff M: (R,W,X) S/U:
 (R,W,X)
Domain0 Next Address      : 0x0000000080200000
Domain0 Next Arg1         : 0x0000000087e00000
Domain0 Next Mode         : S-mode
Domain0 SysReset          : yes
Domain0 SysSuspend        : yes

Boot HART ID              : 0
Boot HART Domain          : root
Boot HART Priv Version    : v1.12
Boot HART Base ISA        : rv64imafdch
Boot HART ISA Extensions  : time,sstc
Boot HART PMP Count       : 16
Boot HART PMP Granularity : 4
Boot HART PMP Address Bits: 54
Boot HART MHPM Count      : 16
Boot HART MIDELEG         : 0x0000000000001666
Boot HART MEDELEG         : 0x0000000000f0b509
```

```
dahyunnist@dahyunnist-ubuntu: ~/os/linux/linux-6.16.9
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$ gdb-multiarch ~/os/linux/l
nux-6.16.9/vmlinux
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from /home/dahyunnist/os/linux/linux-6.16.9/vmlinux...

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for /home/dahyunnist/os/linux/linux-6.16.9/vmlin
x
(No debugging symbols found in /home/dahyunnist/os/linux/linux-6.16.9/vmlinux)
(gdb) target remote :1234
Remote debugging using :1234
0x0000000000001000 in ?? ()
(gdb) b start_kernel
Breakpoint 1 at 0xffffffff80c006b0
(gdb) continue
Continuing.

Breakpoint 1, 0xffffffff80c006b0 in start_kernel ()
(gdb)
```

可以看到GDB在start_kernel函数处触发了断点，QEMU终端在输出OpenSBI信息后停止，未继续输出内核日志。

- 执行命令`quit`，gdb退出：

```
(gdb) quit
A debugging session is active.

        Inferior 1 [process 1] will be detached.

Quit anyway? (y or n) y
Detaching from program: /home/dahyunnist/os/linux/linux-6.16.9/vmlinux, process 1
Ending remote debugging.
[Inferior 1 (process 1) detached]
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$
```

qemu终端继续输出内核日志并进入用户态：

```
[    5.595531] 9pnet: Installing 9P2000 support
[    5.599098] Key type dns_resolver registered
[    9.281773] clk: Disabling unused clocks
[    9.283764] PM: genpd: Disabling unused power domains
[    9.286458] ALSA device list:
[    9.287884]   No soundcards found.
[    9.524384] EXT4-fs (vda): mounted filesystem c3e9bbca-ec22-47f9-a368-187b2117
2fc1 ro with ordered data mode. Quota mode: disabled.
[    9.527401] VFS: Mounted root (ext4 filesystem) readonly on device 254:0.
[    9.541181] devtmpfs: mounted
[    9.813331] Freeing unused kernel image (initmem) memory: 2388K
[    9.818289] Run /sbin/init as init process

Please press Enter to activate this console.
/ #
```

实验结果与分析

分步运行结果如前文所述。经过环境配置，可以成功启动qemu并运行linux内核

```
[   12.279668] EXT4-fs (vda): mounted filesystem c3e9bbca-ec22-47f9-a368-187b21172fc1 ro with orde
red data mode. Quota mode: disabled.
[   12.289398] VFS: Mounted root (ext4 filesystem) readonly on device 254:0.
[   12.306024] devtmpfs: mounted
[   12.642842] Freeing unused kernel image (initmem) memory: 2388K
[   12.648429] Run /sbin/init as init process

Please press Enter to activate this console.
/ # uname -a
Linux (none) 6.16.9 #1 SMP Fri Sep 26 00:21:35 CST 2025 riscv64 GNU/Linux
/ #
```

并且可以成功对内核进行gdb调试

```
dahyunnist@dahyunnist-ubuntu: ~/os/linux/linux-6.16.9
Firmware RW Offset        : 0x40000
Firmware RW Size          : 66 KB
Firmware Heap Offset      : 0x48000
Firmware Heap Size        : 34 KB (total), 2 KB (reserved), 9 KB (used), 22 KB (f
ree)
Firmware Scratch Size     : 4096 B (total), 760 B (used), 3336 B (free)
Runtime SBI Version       : 1.0

Domain0 Name              : root
Domain0 Boot HART         : 0
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Domain0 Region00          : 0x0000000002000000-0x000000000200ffff M: (I,R,W) S/U:
 ()
Domain0 Region01          : 0x0000000080040000-0x000000008005ffff M: (R,W) S/U: (
)
Domain0 Region02          : 0x0000000080000000-0x000000008003ffff M: (R,X) S/U: (
)
Domain0 Region03          : 0x0000000000000000-0xffffffffffffffff M: (R,W,X) S/U:
 (R,W,X)
Domain0 Next Address      : 0x0000000080200000
Domain0 Next Arg1         : 0x0000000087e00000
Domain0 Next Mode         : S-mode
Domain0 SysReset          : yes
Domain0 SysSuspend        : yes

Boot HART ID              : 0
Boot HART Domain          : root
Boot HART Priv Version    : v1.12
Boot HART Base ISA        : rv64imafdch
Boot HART ISA Extensions  : time,sstc
Boot HART PMP Count       : 16
Boot HART PMP Granularity : 4
Boot HART PMP Address Bits: 54
Boot HART MHPM Count      : 16
Boot HART MIDELEG         : 0x0000000000001666
Boot HART MEDELEG         : 0x0000000000f0b509
```

```
dahyunnist@dahyunnist-ubuntu: ~/os/linux/linux-6.16.9
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$ gdb-multiarch ~/os/linux/l
nux-6.16.9/vmlinux
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from /home/dahyunnist/os/linux/linux-6.16.9/vmlinux...

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for /home/dahyunnist/os/linux/linux-6.16.9/vmlin
x
(No debugging symbols found in /home/dahyunnist/os/linux/linux-6.16.9/vmlinux)
(gdb) target remote :1234
Remote debugging using :1234
0x0000000000001000 in ?? ()
(gdb) b start_kernel
Breakpoint 1 at 0xffffffff80c006b0
(gdb) continue
Continuing.

Breakpoint 1, 0xffffffff80c006b0 in start_kernel ()
(gdb)
```

实验中遇到的问题及解决方法

执行实验文档涵盖的操作时未遇到较大问题，但是在实验之外，由于未预留足够的虚拟机磁盘空间，导致虚拟机无法正常启动gdm.service，影响了实验进度。

问题描述：

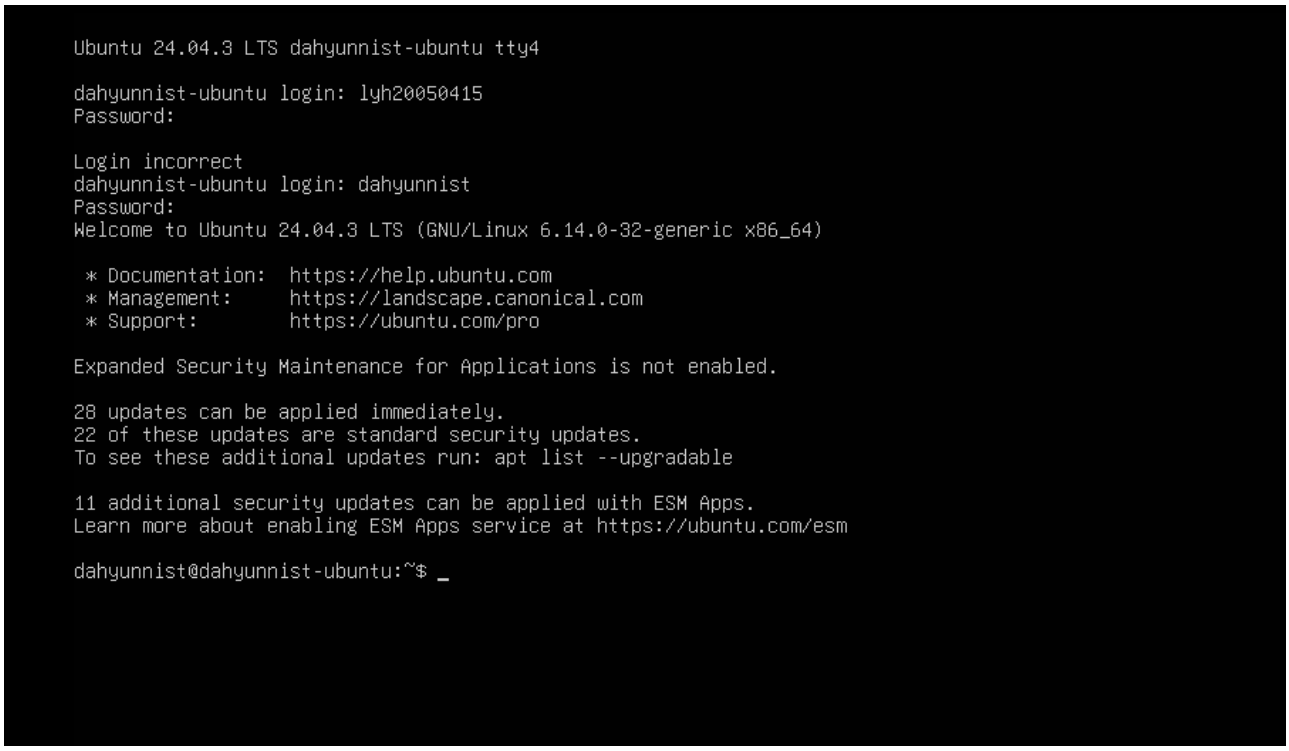
开机后反复提示登录，或是直接出现终端式界面，提示`Failed to start gdm.service - GNOME Display Manager.` 如图：

```
[  OK  ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
[  OK  ] Finished plymouth-quit.service - Terminate Plymouth Boot Screen.
[   59.479169] Bluetooth: hci0: unexpected cc 0x0c12 length: 2 < 3
[   59.479193] Bluetooth: hci0: Opcode 0x0c12 failed: -38
[FAILED] Failed to start gdm.service - GNOME Display Manager.
-
```

解决方案

1. 给虚拟机磁盘扩展空间：参考了csdn上的教程

2. 启动虚拟机，由于还未分区，所以此时仍然会出现终端式界面并提示`Failed to start gdm.service - GNOME Display Manager.`，此时按下`Ctrl+Alt+F4`进入命令行界面(24.04版本结尾是4所以按`f4`):



3. 执行以下命令查看磁盘使用情况和分区状况

```
# 查看磁盘空间使用
df -h
# 查看磁盘分区表
lsblk
```

结果如图：

```
dahyunnist@dahyunnist-ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           387M  1.6M  386M   1% /run
/dev/sda2        20G   19G     0 100% /
tmpfs           1.9G     0  1.9G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           1.9G     0  1.9G   0% /run/qemu
tmpfs           387M   84K  387M   1% /run/user/1000
dahyunnist@dahyunnist-ubuntu:~$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0      2:0    1     4K  0 disk
loop0    7:0    0  73.9M  1 loop /snap/core22/2082
loop1    7:1    0     4K  1 loop /snap/bare/5
loop2    7:2    0  73.9M  1 loop /snap/core22/2133
loop3    7:3    0 247.1M  1 loop /snap/firefox/6933
loop4    7:4    0 247.1M  1 loop /snap/firefox/6836
loop5    7:5    0  11.1M  1 loop /snap/firmware-updater/167
loop6    7:6    0   516M  1 loop /snap/gnome-42-2204/202
loop7    7:7    0 516.2M  1 loop /snap/gnome-42-2204/226
loop8    7:8    0  10.8M  1 loop /snap/snap-store/1270
loop9    7:9    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop10   7:10   0  49.3M  1 loop /snap/snapd/24792
loop11   7:11   0  50.8M  1 loop /snap/snapd/25202
loop12   7:12   0   576K  1 loop /snap/snapd-desktop-integration/315
sda      8:0    0    40G  0 disk
├─sda1   8:1    0     1M  0 part
└─sda2   8:2    0    20G  0 part /
sr0     11:0    1  95.3M  0 rom
sr1     11:1    1   5.9G  0 rom
dahyunnist@dahyunnist-ubuntu:~$
```

可知`/dev/sda`磁盘空间有40G，但是`sda2`只有20G，且处于`100%`使用状态

4. 查看未分配空间的具体位置：

```
fdisk -l /dev/sda
```

5. 使用`fdisk`扩展`/dev/sda2`分区：

```
sudo fdisk /dev/sda
```

会出现`Command (m for help):`，随后：

1. 删除现有根分区：输入`d`回车，提示`Partition number (1,2, default 2)`，输入`2`回车(仅删除分区表中的`/dev/sda2`记录，不会删除实际数据，但需要后续正确重建分区)
2. 重建根分区：
   - 输入`n`回车（新建分区，若提示`Partition type`直接回车默认`Primary`）
   - 提示`Partition number (2-4, deault 2):`，输入`2`回车
   - 提示`First sector (xxxxx-xxxxx, default xxxxx):`，**输入此前第4步查看到的`sda2`起始位置**
   - 提示`Last sector`，回车默认扩展到磁盘末尾
   - 提示`Do you want to remove the signature? [Y]es/[N]o:`，输入`n`回车，保留ext4签名（`ext4 signature`是文件系统的身份标识，记录了`/dev/sda2`是一个已格式化的ext4分

区（包含系统和数据），如果删除，则相当于格式化分区表记录，后续无法识别分区中的数据，即数据彻底丢失）

- 提示`Command (m for help):`输入`w`回车，保存并退出

3. 完整流程如图：

```
dahyunnist@dahyunnist-ubuntu:~$ sudo fdisk -l /dev/sda
GPT PMBR size mismatch (41943039 != 83886079) will be corrected by write.
Disk /dev/sda: 40 GiB, 42949672960 bytes, 83886080 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: BF2E446F-F2CB-4E0C-B474-7F5D0E23FF4C

Device     Start      End  Sectors Size Type
/dev/sda1   2048     4095     2048   1M BIOS boot
/dev/sda2   4096 41940991 41936896  20G Linux filesystem
dahyunnist@dahyunnist-ubuntu:~$ sudo fdisk /dev/sda

Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

GPT PMBR size mismatch (41943039 != 83886079) will be corrected by write.
This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.


Command (m for help): d
Partition number (1,2, default 2): 2

Partition 2 has been deleted.

Command (m for help): n
Partition number (2-128, default 2): 2
First sector (4096-83886046, default 4096): _
```

```
Command (m for help): d
Partition number (1,2, default 2): 2

Partition 2 has been deleted.

Command (m for help): n
Partition number (2-128, default 2): 2
First sector (4096-83886046, default 4096): 4096
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4096-83886046, default 83884031):

Created a new partition 2 of type 'Linux filesystem' and of size 40 GiB.
Partition #2 contains a ext4 signature.

Do you want to remove the signature? [Y]es/[N]o:
```

```
Do you want to remove the signature? [Y]es/[N]o: n

Command (m for help): w

The partition table has been altered.
Syncing disks.

dahyunnist@dahyunnist-ubuntu:~$ sudo e2fsck -fy /dev/sda2
e2fsck 1.47.0 (5-Feb-2023)
/dev/sda2 is mounted.
e2fsck: Cannot continue, aborting.


dahyunnist@dahyunnist-ubuntu:~$
```

6. 将ext4文件系统扩展到分区最大容量

```
sudo resize2fs /dev/sda2
```

并检查：

```
df -h /
```

结果如图：

```
dahyunnist@dahyunnist-ubuntu:~$ sudo mount -o remount,ro /
mount: /: mount point is busy.
       dmesg(1) may have more information after failed mount system call.
dahyunnist@dahyunnist-ubuntu:~$ sudo resize2fs /dev/sda2
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/sda2 is mounted on /; on-line resizing required
old_desc_blocks = 3, new_desc_blocks = 5
The filesystem on /dev/sda2 is now 10484992 (4k) blocks long.

dahyunnist@dahyunnist-ubuntu:~$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        40G   19G   19G  50% /
```

可以看到扩展成功

7. 重启GDM服务：

```
sudo systemctl restart gdm.service
```

可以重新进入图形化登录界面，正常开机。

思考题与心得体会

1. 使用`riscv64-linux-gnu-gcc`编译单个`.c`文件
   - 新建一个计算阶乘的test.c文件：

```
dahyunnist@dahyunnist-ubuntu:~/os$ ls
linux  os-25fall  test.c
dahyunnist@dahyunnist-ubuntu:~/os$ cat test.c
#include <stdio.h>
int fact(int n){
    if(n == 1) return 1;
    else return n * fact(n-1);
}

int main(void)
{
    printf("The factorial of 5 is %d\n", fact(5));
    return 0;
}dahyunnist@dahyunnist-ubuntu:~/os$
```

   - 执行命令`riscv64-linux-gnu-gcc -o test test.c`进行编译

2. 使用 `riscv64-linux-gnu-objdump` 反汇编 1 中得到的编译产物

- 执行命令`riscv64-linux-gnu-objdump -d test > test.dis`进行反编译，并查看反编译结果：

```
dahyunnist@dahyunnist-ubuntu:~/os$ riscv64-linux-gnu-gcc -o test test.c
dahyunnist@dahyunnist-ubuntu:~/os$ riscv64-linux-gnu-objdump -d test > test.dis
dahyunnist@dahyunnist-ubuntu:~/os$ ls
linux  os-25fall  test  test.c  test.dis
dahyunnist@dahyunnist-ubuntu:~/os$ cat test.dis

test:     file format elf64-littleriscv


Disassembly of section .plt:

0000000000000570 <.plt>:
 570:   00002397                auipc   t2,0x2
```

其中主要部分(`fact`函数和`main`函数)为：

```
0000000000000694 <fact>:
 694:   1101                    addi    sp,sp,-32
 696:   ec06                    sd      ra,24(sp)
 698:   e822                    sd      s0,16(sp)
 69a:   1000                    addi    s0,sp,32
 69c:   87aa                    mv      a5,a0
 69e:   fef42623                sw      a5,-20(s0)
 6a2:   fec42783                lw      a5,-20(s0)
 6a6:   0007871b                sext.w  a4,a5
 6aa:   4785                    li      a5,1
 6ac:   00f71463                bne     a4,a5,6b4 <fact+0x20>
 6b0:   4785                    li      a5,1
 6b2:   a831                    j       6ce <fact+0x3a>
 6b4:   fec42783                lw      a5,-20(s0)
 6b8:   37fd                    addiw   a5,a5,-1
 6ba:   2781                    sext.w  a5,a5
 6bc:   853e                    mv      a0,a5
 6be:   fd7ff0ef                jal     694 <fact>
 6c2:   87aa                    mv      a5,a0
 6c4:   fec42703                lw      a4,-20(s0)
 6c8:   02f707bb                mulw    a5,a4,a5
 6cc:   2781                    sext.w  a5,a5
 6ce:   853e                    mv      a0,a5
 6d0:   60e2                    ld      ra,24(sp)
 6d2:   6442                    ld      s0,16(sp)
 6d4:   6105                    addi    sp,sp,32
 6d6:   8082                    ret
```

```
00000000000006d8 <main>:
 6d8:   1141                    addi    sp,sp,-16
 6da:   e406                    sd      ra,8(sp)
 6dc:   e022                    sd      s0,0(sp)
 6de:   0800                    addi    s0,sp,16
 6e0:   4515                    li      a0,5
 6e2:   fb3ff0ef                jal     694 <fact>
 6e6:   87aa                    mv      a5,a0
 6e8:   85be                    mv      a1,a5
 6ea:   00000517                auipc   a0,0x0
 6ee:   02650513                addi    a0,a0,38 # 710 <_IO_stdin_used+0x8>
 6f2:   eafff0ef                jal     5a0 <printf@plt>
 6f6:   4781                    li      a5,0
 6f8:   853e                    mv      a0,a5
 6fa:   60a2                    ld      ra,8(sp)
 6fc:   6402                    ld      s0,0(sp)
 6fe:   0141                    addi    sp,sp,16
 700:   8082                    ret
```

3. 调试 Linux 时:

    1. 尝试使用 GDB 的各项命令（如 `backtrace`, `finish`, `frame`, `info`, `break`, `display`, `next`, `layout` 等）。 尝试结果如下:

```
(gdb) backtrace
#0  0xffffffff80c006b0 in start_kernel ()
#1  0xffffffff80001172 in _start_kernel ()
Backtrace stopped: frame did not save the PC
```

```
(gdb) info registers
ra             0xffffffff80001172       0xffffffff80001172 <_start_kernel+154>
sp             0xffffffff81803ee0       0xffffffff81803ee0
gp             0xffffffff81a19d88       0xffffffff81a19d88 <__SCK__tp_func_cgroup
_rstat_cpu_unlock>
tp             0xffffffff8180e0c0       0xffffffff8180e0c0 <init_task>
t0             0x81005000       2164281344
t1             0xffffffff80c00172       -2134900366
t2             0x1000   4096
fp             0x80046ef0       0x80046ef0
s1             0x1      1
a0             0xff1bffffec00000       -64176294711001088
a1             0x0      0
a2             0xffffffff80c4f3d8       -2134576168
a3             0xffffffff80c0768e       -2134870386
a4             0xffffffff80c06608       -2134874616
a5             0xffffffff80c0760e       -2134870514
a6             0xffffffff80c0778e       -2134870130
a7             0xffffffff80c065b6       -2134874698
s2             0x0      0
s3             0x80200000       2149580800
s4             0xfffffffffffffff3       -13
s5             0x87e00000       2279604224
s6             0x8000000a00006800       -9223371993905076224
s7             0x80040040       2147745856
s8             0x2000   8192
s9             0x800426e0       2147755744
s10            0x0      0
s11            0x0      0
t3             0x0      0
t4             0x7      7
t5             0xf      15
t6             0x27     39
pc             0xffffffff80c006b0       0xffffffff80c006b0 <start_kernel>
```

```
(gdb) display /i $pc
1: x/i $pc
=> 0xffffffff80c006b0 <start_kernel>:   addi    sp,sp,-96
```

```
(gdb) step
Single stepping until exit from function arch_cpu_idle,
which has no line number information.
0xffffffff80b211bc in default_idle_call ()
1: x/i $pc
=> 0xffffffff80b211bc <default_idle_call+30>:
    jal 0xffffffff80b2099c <ct_idle_exit>
```

2. 在 GDB 中查看汇编代码（不使用任何插件的情况下）
3. 在 `0x80000000` 处下断点
4. 查看所有已下的断点
5. 在 `0x80200000` 处下断点
6. 清除 `0x80000000` 处的断点
7. 继续运行直到触发 `0x80200000` 处的断点
8. 单步调试一次

9. 退出 QEMU 2-4完整过程如下：

```
(gdb) break *0x80000000
Breakpoint 1 at 0x80000000
(gdb) info breakpoints
Num     Type           Disp Enb Address            What
1       breakpoint     keep y   0x0000000080000000
(gdb) break *0x80200000
Breakpoint 2 at 0x80200000
(gdb) delete break 1
(gdb) info breakpoints
Num     Type           Disp Enb Address            What
2       breakpoint     keep y   0x0000000080200000
(gdb) continue
Continuing.

Breakpoint 2, 0x0000000080200000 in ?? ()
(gdb) nexti
0x0000000080200002 in ?? ()
(gdb) layout asm
(gdb) quit
A debugging session is active.

        Inferior 1 [process 1] will be detached.

Quit anyway? (y or n) y
Detaching from program: /home/dahyunnist/os/linux/linux-6.16.9/vmlinux, process 1
Ending remote debugging.
[Inferior 1 (process 1) detached]
```

4. 使用 `make` 工具清除 Linux 的构建产物：

```
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$ make clean
  CLEAN   drivers/firmware/efi/libstub
  CLEAN   drivers/gpu/drm/radeon
  CLEAN   drivers/scsi
  CLEAN   drivers/tty/vt
  CLEAN   init
  CLEAN   kernel
  CLEAN   lib/raid6
  CLEAN   lib
  CLEAN   security/apparmor
  CLEAN   security/selinux
  CLEAN   usr
  CLEAN   .
  CLEAN   modules.builtin modules.builtin.modinfo .vmlinux.export.c
dahyunnist@dahyunnist-ubuntu:~/os/linux/linux-6.16.9$ make mrproper
  CLEAN   scripts/basic
  CLEAN   scripts/dtc
  CLEAN   scripts/kconfig
  CLEAN   scripts/mod
  CLEAN   scripts/selinux/mdp
  CLEAN   scripts
  CLEAN   include/config include/generated .config .version Module.symvers
```

5. `vmlinux` 和 `Image` 的关系和区别是什么？

- 关系：`Image`是`vmlinux`在Linux内核编译时经过`objcopy`转换处理后形成的二进制内核映像
- 区别：
  - `vmlinux`是`elf`格式，而`Image`是二进制格式
  - `vmlinux`中除二进制代码外还包含完整符号和调试信息，而`Image`中只包含二进制数据

- `vmlinux`可用于调试和定位内核问题，不能直接引导Linux系统启动；而`Image`正相反